

**VALIDATION REPORT:
PHOTON ATTENUATION SIMULATOR
(V1.11)**

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1. EXECUTIVE SUMMARY

This report documents validation of the Shielding Attenuation Simulator V1.11. Version 1.11 introduces a reproducible scenario and concentric spherical geometry layer above the previously validated V1.10 deterministic shielding and optimization system. The release replaces hard coded example inputs with validated source, geometry, target, constraint, and optimization specifications that can be saved as human-readable JSON and reconstructed without loss of information.

The Beer-Lambert attenuation model, NIST XCOM interpolation, inverse-square point-source model, isotope calculations, G-P exposure buildup implementation, minimum thickness solver, material comparison, constraint filtering, objective ranking, Pareto analysis, and engineering visualization are unchanged from V1.10. V1.11 therefore does not require a new derivation or independent revalidation of those physics and optimization equations. The V1.10 validation report remains the controlling record for the underlying numerical models.

A focused V1.11 test program validates the new geometry and scenario architecture: shell radius derivation, source and material key normalization, scenario level consistency rules, JSON serialization and loading, schema rejection, malformed file handling, runtime source and material adapters, and reproduction of the official constrained Cs-137 design result. The complete validation runner produced 319 PASS assertions and ended with "All validation tests passed." Of these, 46 assertions directly exercise V1.11 behavior.

The official JSON scenario reproduced the documented V1.10 result: Lead was selected from nine candidate materials at 7.088213 cm with a mass per area of 80.380339 g/cm². An independent OpenMC environment smoke test also completed successfully using OpenMC 0.15.3 in Docker, confirmed access to the continuous energy data library, executed a fixed source photon calculation, wrote a statepoint, and read tally results through the Python API. This smoke test verifies environment readiness only; it is not yet a deterministic versus Monte Carlo benchmark.

2. SCOPE AND VALIDATION METHOD

2.1 Previously Validated Calculations

V1.11 routes scenario inputs into the existing V1.09/V1.10 calculation pathways. It does not alter attenuation coefficients, source spectra, buildup factors, minimum thickness methods, constraint logic, objective ranking, or plotting calculations. Regression preservation is therefore demonstrated by retaining all 273 previously documented PASS assertions and reproducing the official Lead result.

2.2 New V1.11 Components

- `geometry_models.py` defines layer specifications, derived spherical shells, and radial geometry validation.
- `scenario_models.py` defines serializable sources, deterministic settings, and the complete `ShieldingScenario` model.
- `scenario_io.py` provides typed JSON conversion, save/load, schema checks, and readable error handling.
- `scenario_runner.py` resolves runtime sources and materials, then delegates to the existing optimizer.
- The official JSON case, one command runner, architecture documentation, and independent OpenMC smoke test complete the reproducibility workflow.

3. TEST CASE RESULTS

3.1 Automated Validation Summary

Validation Area	Acceptance Criterion	Result
Spherical geometry	Layer keys normalize; shell thicknesses and inner/outer radii derive correctly; invalid radial geometry is rejected.	PASS
Source and settings models	Manual and isotope specifications preserve physical inputs; invalid energy, activity, and search limits are rejected.	PASS
Scenario consistency	Calculation mode, candidates, target, constraints, objective, weights, fixed layers, and radial search space are validated.	PASS
JSON serialization	Explicit source, geometry, and target types are preserved; dictionary and file round trips reconstruct equivalent objects.	PASS
JSON error handling	Unsupported schema versions, malformed JSON, missing fields, and invalid nested values fail with readable errors.	PASS
Runtime adapters	Source specifications and material keys resolve to the correct existing runtime objects without changing their values or order.	PASS
Reference regression	The JSON scenario evaluates nine candidates and reproduces the selected Lead thickness and mass per area.	PASS
Earlier regression suite	All V1.10 and earlier attenuation, buildup, source, design, optimization, and plotting tests remain passing.	PASS

The current runner contains 319 PASS assertions. The V1.10 report documented 273 PASS assertions; V1.11 adds 46 focused assertions while preserving the complete earlier regression suite.

3.2 Geometry and Scenario Model Validation

A two-layer reference geometry used a 2.0 cm source cavity, 5.0 cm of Lead, 1.0 cm of Copper, and a 100 cm evaluation radius. The geometry model correctly derived a total shield thickness of 6.0 cm, an outer shield radius of 8.0 cm, a Lead shell spanning 2.0-7.0 cm, and a Copper shell spanning 7.0-8.0 cm. Tests also confirmed rejection of nonpositive layer thickness and an evaluation radius located at or inside the outer shield surface.

Scenario level validation prevents ambiguous combinations. Material optimization scenarios require at least one unique candidate material, reject fixed shield layers, require weights only for

balanced optimization, and restrict the numerical thickness search to the available radial distance between the source cavity and evaluation location.

3.3 JSON Serialization and Loading

The JSON format uses explicit type fields for sources, geometry, and targets. Derived shell radii are excluded because they are reconstructed from the cavity radius and ordered thicknesses. The schema 1.0 loader rejects unsupported versions, malformed JSON, missing fields, and invalid nested values; the official save/load round trip produced an equivalent ShieldingScenario object.

4. OFFICIAL V1.11 ENGINEERING CASE

4.1 Input Scenario

Parameter	Value
Scenario ID	cs137_v110_reference
Source	Cs-137 isotope source
Activity	$3.70 \cdot 10^{10}$ Bq (1.00 Ci)
Geometry	Concentric spherical; 0 cm source cavity; 100 cm evaluation radius; no fixed layers
Candidate materials	Nine materials in the Python material library
Design target	Flux target: 100 photons/cm ² /s
Engineering constraints	Maximum thickness 20 cm; maximum mass per area 120 g/cm ²
Optimization objective	Minimum mass per area
Deterministic settings	G-P exposure buildup requested; calculation search limit 100 cm

Execution through load_scenario() and run_scenario() evaluated nine candidates and selected Lead. The required thickness was 7.088213273 cm and the mass per area was 80.380338518 g/cm². These values agree with the V1.10 documented values within the established validation tolerances. The result also preserved the selection explanation that Lead has the lowest mass per area among the two eligible candidates.

4.2 OpenMC Environment Smoke Test

Item	Observed Result
Environment	Official OpenMC Docker image; repository mounted at /workspace; OpenMC executable and Python API version 0.15.3
Continuous energy data	/root/nndc_hdf5/cross_sections.xml found and recognized by the Python API
Source and geometry	661.657 keV isotropic photon point source; 1 cm cavity; 1 cm Lead shell; 10 cm vacuum boundary
Run settings	Fixed source; photon transport enabled; 5 batches · 2,000 particles; seed 12,345
Generated artifacts	model.xml, statepoint.5.h5, summary.h5, and tallies.out
Statepoint checks	Fixed source mode, photon transport enabled, five realizations, and named tally retrieved
Raw tally	Mean 3.35679731; standard deviation 0.028220853
Leakage / runtime	0.41610 +/- 0.00363; elapsed time 0.45296 s
Result	PASS - environment, transport, statepoint, and Python reading pathway verified

The raw outer void cell flux tally is not a detector response, an air-kerma result, or a quantity matched to the deterministic model. It is reported only as environment verification, not as OpenMC benchmark validation.

5. V1.11 ARCHITECTURE

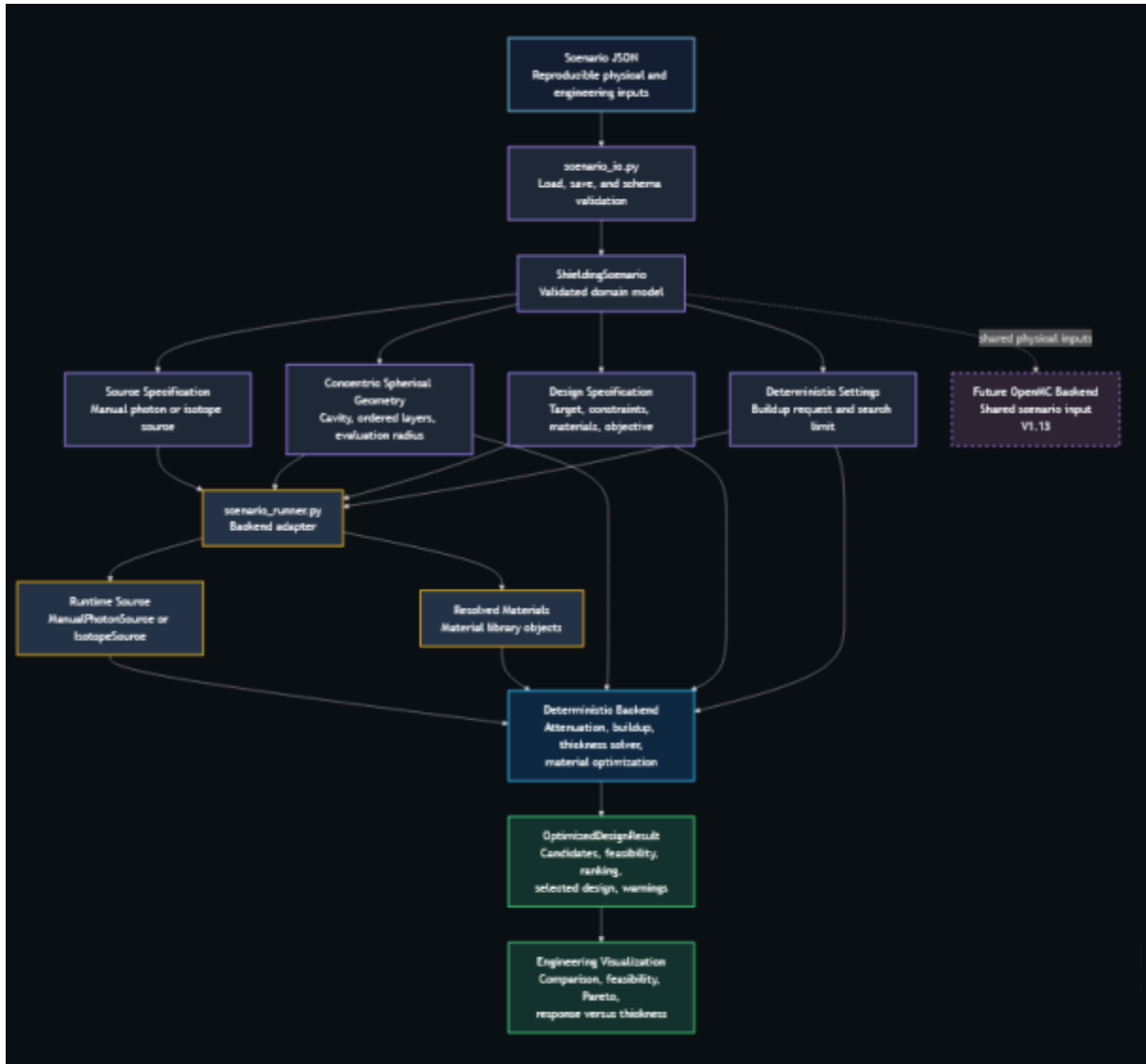


Figure 1. V1.11 scenario architecture. JSON parsing and backend adaptation are separated from the previously validated deterministic calculation modules. The dashed OpenMC branch is planned for V1.13 and is not implemented by the V1.11 scenario runner.

The architecture makes ShieldingScenario the authoritative problem definition while preserving separation of responsibilities. scenario_io.py handles files and schema reconstruction; scenario_models.py and geometry_models.py enforce domain invariants; scenario_runner.py converts portable specifications into the runtime source and material objects expected by the existing optimizer. This design reduces the risk that future deterministic and OpenMC calculations silently use different source, geometry, target, or material inputs.

6. ASSUMPTIONS AND LIMITATIONS

1. Previously Validated Physics Reused: V1.11 validates scenario architecture, spherical geometry bookkeeping, serialization, adapters, and environment readiness rather than revalidating the V1.10 attenuation and optimization equations
2. Geometry Specification Versus Transport Geometry: ConcentricSphericalGeometry defines physical radii and layer order, but the current deterministic backend still uses the evaluation radius as the existing center-to-detector distance. It does not yet calculate full three-dimensional transport or total spherical shield mass.
3. Material Optimization Mode Only: The V1.11 scenario runner supports material_optimization. Fixed shield layers are rejected in this mode to prevent them from being silently ignored or ambiguously combined with optimizer generated thicknesses.
4. Schema Version 1.0 Only: The JSON loader intentionally rejects unknown schema versions. Migration logic for future scenario formats has not yet been implemented.
5. No Run Manifest or Scenario Hash: V1.11 provides reproducible JSON inputs but does not yet create cryptographic scenario fingerprints, execution manifests, or archived result bundles.
6. OpenMC Not Yet Integrated: The smoke test script is intentionally independent of ShieldingScenario and scenario_runner.py. Shared scenario to OpenMC model construction begins in V1.13.
7. No Matched OpenMC Observable: The raw outer void cell flux tally was used only to prove execution and statepoint access. Its spatial averaging and per-source particle normalization are not directly comparable to the deterministic detector flux.
8. Docker Based Environment: The verified OpenMC pathway uses the official Docker image and its included continuous energy data. Reproducibility on a native Windows Python environment was not evaluated.
9. Conceptual Engineering Use: The simulator and scenario files are educational and conceptual engineering tools. V1.11 does not certify shielding performance or replace detailed transport analysis and regulatory review.

7. CONCLUSION

The Shielding Attenuation Simulator V1.11 successfully introduces a reproducible scenario and concentric spherical geometry foundation while preserving all previously validated deterministic behavior. The release converts portable JSON inputs into validated domain objects, rejects inconsistent scenario combinations, resolves runtime sources and materials through stable keys, and delegates calculations to the existing shielding optimizer without duplicating physics logic.

The focused validation program added 46 V1.11 assertions. Together with the retained regression suite, the complete runner produced 319 PASS assertions and no failures. The official JSON scenario reproduced the nine material Cs-137 problem and selected Lead at 7.088213 cm with 80.380339 g/cm² mass per area.

The OpenMC smoke test independently confirmed that the repository can be mounted into an OpenMC 0.15.3 Docker environment, access continuous energy data, execute fixed source photon transport, write a statepoint, and recover tally values through Python. V1.11 therefore completes the reproducibility and environment foundation required for V1.12 response quantity clarification and the matched OpenMC benchmark planned for V1.13.

References

- [1] Y. Harima, Y. Sakamoto, S. Tanaka, and M. Kawai, "Validity of the Geometric-Progression Formula in Approximating Gamma-Ray Buildup Factors," *Nuclear Science and Engineering*, vol. 94, no. 1, pp. 24–35, Sep. 1986. doi:10.13182/nse86-a17113
- [2] D. K. Trubey, "New gamma-ray buildup factor data for point kernel calculations: ANS-6.4.3 standard reference data," IAEA, <https://inis.iaea.org/records/0arzw-ez976/files/20014493.pdf> (accessed Jul. 10, 2026).
- [3] Angstrom Sciences, "Density of elements chart – Angstrom Sciences elements density table," Angstrom Sciences, <https://angstromsciences.com/density-elements-chart> (accessed Mar. 26, 2026).
- [4] J. Colledge, M. Wilczek, and C. Raymond, "Inverse square law | radiology reference article," Radiopaedia, <https://radiopaedia.org/articles/inverse-square-law?lang=us> (accessed Mar. 26, 2026).
- [5] "Shielding Radiation Alphas, Betas, Gammas and Neutrons," U.S. NRC, <https://www.nrc.gov/docs/ML1122/ML11229A721.pdf> (accessed Mar. 26, 2026).
- [6] M. J. Berger et al., "XCOM: Photon Cross Sections Database," NIST, <https://dx.doi.org/10.18434/T48G6X> (accessed Mar. 26, 2026).
- [7] National Nuclear Data Center, "NuDat 3 Glossary," Brookhaven National Laboratory, <https://www.nndc.bnl.gov/nudat3/guide/glossary.html> (accessed Jul. 11, 2026).
- [8] International Atomic Energy Agency, "LiveChart of the Nuclides," IAEA Nuclear Data Section, <https://nds.iaea.org/relnsd/vcharthtml/VChartHTML.html> (accessed Jul. 11, 2026).
- [9] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," *Computing in Science & Engineering*, vol. 9, no. 3, pp. 90–95, 2007. doi:10.1109/MCSE.2007.55 (accessed Jul. 20, 2026).
- [10] OpenMC Project, "Installation and Configuration," OpenMC Documentation, <https://docs.openmc.org/en/latest/usersguide/install.html> (accessed July 23, 2026).
- [11] OpenMC Project, "Execution Settings - Photon Transport," OpenMC Documentation, <https://docs.openmc.org/en/stable/usersguide/settings.html> (accessed July 23, 2026).

- [12] OpenMC Project, "openmc.StatePoint," OpenMC Python API Documentation,
<https://docs.openmc.org/en/stable/pythonapi/generated/openmc.StatePoint.html> (accessed July 23, 2026).
- [13] OpenMC Project, "State Point File Format," OpenMC Documentation,
https://docs.openmc.org/en/stable/io_formats/statepoint.html (accessed July 23, 2026).